

Case Study: Predicting Student Pass/Fail Using a Neural Network

1. Background

A college wants to build a simple AI model that can look at a student's performance and predict whether that student is likely to Pass or Fail. This is a common beginner-level machine learning problem because it does not need huge amounts of data or a very complex model. The college has collected data for 500 students, and for each student it has recorded three simple details:

- Study Hours – how many hours the student studies per week
- Attendance – the percentage of classes the student has attended
- Assignment Marks – the marks scored in assignments

Using these three details, the model has to make one decision: will the student Pass or Fail? This kind of problem, where the answer is one of two options, is called a binary classification problem.

2. Which Architecture Is More Suitable: Single Hidden Layer or Multiple Hidden Layers?

For this problem, a Single Hidden Layer neural network is the better and more practical choice. Here is why.

The relationship between study hours, attendance, assignment marks and the final result (pass or fail) is fairly straightforward. A student who studies more, attends more classes, and scores well in assignments is naturally more likely to pass. There are no hidden layers of complexity like there would be in problems such as image recognition or speech recognition, where the model needs to learn many detailed patterns, edges, shapes, or sounds.

Multiple hidden layers (also called deep neural networks) are useful when the data is very large, unstructured, or has complicated patterns, for example thousands of images or long pieces of text. Since this dataset only has 500 records and just three simple numeric inputs, adding more hidden layers would not really improve accuracy. Instead, it would make the model heavier, slower to train, and more likely to overfit, meaning it may memorize the training data instead of actually learning the general pattern.

So, for a small and simple dataset like this one, a single hidden layer is enough to capture the relationship between the inputs and the output.

3. Identifying the Layers

Every neural network is made up of three basic types of layers. Here is how they apply to this problem:

Layer	What It Contains	Role in This Problem
Input Layer	3 neurons: Study Hours, Attendance, Assignment Marks	Receives the raw student data
Hidden Layer	5 neurons (in this example)	Learns the pattern connecting inputs to the result
Output Layer	1 neuron	Gives the final prediction: Pass or Fail

4. Neural Network Architecture Diagram (1 Hidden Layer, 5 Neurons)

Below is a simple diagram showing how the three input neurons connect to the five hidden neurons, which then connect to the single output neuron. Every neuron in one layer is connected to every neuron in the next layer, this is called a fully connected network.

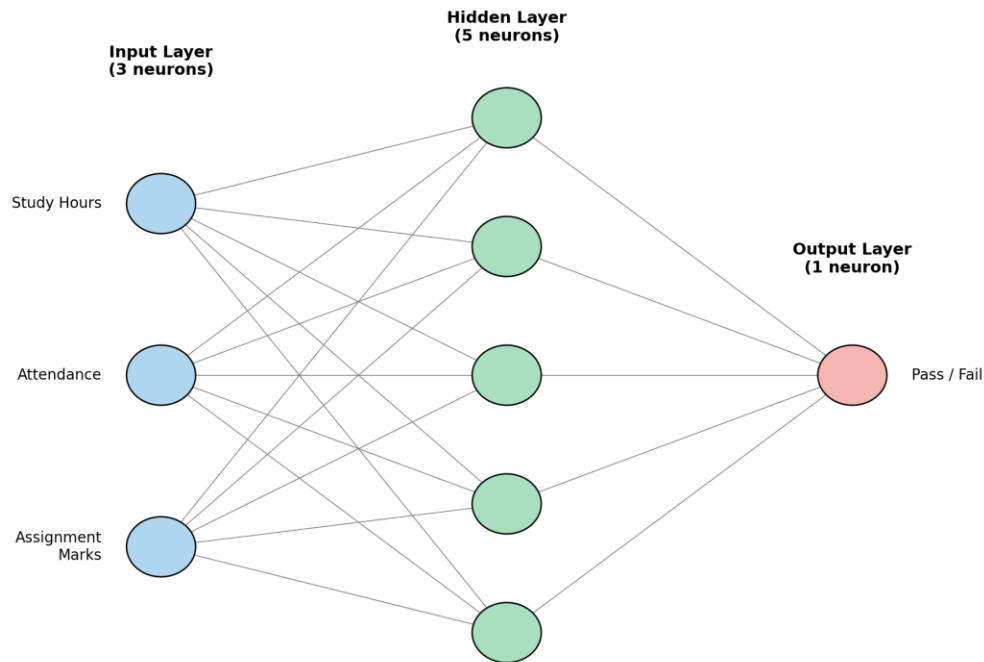


Figure 1: A single hidden layer neural network with 3 input neurons, 5 hidden neurons, and 1 output neuron.

5. Why Is a Single Hidden Layer Sufficient for This Problem?

There are a few simple reasons why one hidden layer is enough here, instead of using a deep network with many layers:

- **Small dataset:** With only 500 records, a deep network does not have enough data to justify its extra complexity.
- **Simple relationship:** The connection between study hours, attendance, marks and the pass/fail outcome is direct and not deeply layered.
- **Faster training:** A single hidden layer trains much faster and needs less computing power.
- **Avoids overfitting:** Fewer layers means the model is less likely to just memorize the training data and more likely to generalize well to new students.
- **Proven by theory:** According to the Universal Approximation Theorem, a neural network with just one hidden layer (with enough neurons) can approximate almost any simple function, which is exactly what this problem needs.

6. Conclusion

For a small, structured dataset like this student performance data, a single hidden layer neural network with a handful of neurons (such as 5) is the most suitable choice. It is simple to build, quick to train, easy to understand, and accurate enough for the task of predicting whether a student will pass or fail. Multiple

hidden layers would only be needed if the data became much larger or far more complex, such as raw images, audio, or text.

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